



Modeling with Exponential Equations and Expressions

Exponential equations and expressions can be used to represent quantities that grow or decay by a fixed percent at regular intervals.

Identifying Growth, Decay, and Rates of Change

An exponential function has the form $y = a(b)^x$, where a is any nonzero real number and b is the base of the function such that $b > 0$ and $b \neq 1$. When $a > 0$ and $b > 1$, the function increases or grows exponentially. When $a > 0$ and $0 < b < 1$, the function decreases or decays exponentially.

Definition

Exponential growth occurs when a quantity increases by the same percent at regular intervals over a given period of time. This situation is modeled by the equation

$$y = A(1 + r)^t$$

where A is the initial amount, r is the rate of change expressed as a decimal, and t represents time.

For example, if \$2500 is invested at an annual interest rate of 3% for t years, then the equation $y = 2500(1 + 0.03)^t = 2500(1.03)^t$ represents the value of the investment after t years.

Definition

Exponential decay occurs when a quantity decreases by the same percent at regular intervals over a given period of time. This situation is modeled by the equation

$$y = A(1 - r)^t$$

where A is the initial amount, r is the rate of change expressed as a decimal, and t represents time.

For example, if a bicycle tire contains 1400 in³ of air and loses 2.5% of its air each day, then the equation $y = 1400(1 - 0.025)^t$ or $y = 1400(0.975)^t$ represents the amount of air in the tire after t days.

EXAMPLE 1

Determine whether each equation represents exponential growth or decay. Then find the rate of growth or decay.

A $y = 250(0.86)^x$

SOLUTION

Transform the equation into the form $y = A(1 + r)^t$ or $y = A(1 - r)^t$.

$$\begin{aligned}y &= 250(0.86)^x \\y &= 250(1 - 0.14)^x\end{aligned}$$

The equation can be rewritten in the form $y = A(1 - r)^x$, where $A = 250$ and $r = 0.14$. Because the rate is subtracted from 1, the equation represents exponential decay.

$$0.14 = \frac{14}{100} = 14\% \quad \text{Write the rate as a percent.}$$

The rate of decay is a 14% decrease every x units of time.

B $y = 1.44^{\frac{t}{2}}$

SOLUTION

Use the properties of exponents to rewrite the equation with only t in the exponent.

$$y = 1.44^{\frac{t}{2}}$$

$$y = 1.44^{\frac{1}{2} \cdot t} \quad \text{Rewrite the exponent.}$$

$$y = \left(1.44^{\frac{1}{2}}\right)^t \quad \text{Power of a Power Rule}$$

$$y = (1.2)^t \quad \text{Simplify.}$$

Transform the exponential equation into the form $y = A(1 + r)^t$
or $y = A(1 - r)^t$.

$$y = (1.2)^t$$

$$y = (1 + 0.2)^t$$

The equation can be rewritten in the form $y = A(1 + r)^t$, where $A = 1$ and $r = 0.2$. Because the rate is added to 1, the equation represents exponential growth.

$$0.20 = \frac{20}{100} = 20\% \quad \text{Write the rate as a percent.}$$

The rate of growth is a 20% increase every t units of time. ■

► **Think About It** When an exponential model is in the form $y = a(b)^x$, you can tell at a glance whether it represents exponential growth or decay by examining b . If $0 < b < 1$, the model represents exponential decay. If $b > 1$, the model represents exponential growth.

Modeling with Exponential Equations

You can use exponential equations to model and solve problems.

EXAMPLE 2

The number of visitors to a state park grows at an annual rate of about 1.69%. This year, the number of visitors was 44,505.

A Write an exponential growth equation to model the situation.

SOLUTION

The initial amount is $A = 44,505$.

The annual growth rate is 1.69%. Write this rate as a decimal.

$$r = 1.69\% = \frac{1.69}{100} = 0.0169$$

Substitute the values of A and r into the equation $y = A(1 + r)^t$ and simplify.

$$\begin{aligned} y &= 44,505(1 + 0.0169)^t \\ &= 44,505(1.0169)^t \end{aligned}$$

The exponential equation $y = 44,505(1.0169)^t$ models the number of annual visitors to the park after t years.

B According to the model, about how many visitors are expected 10 years from now?

SOLUTION

Substitute $t = 10$ in the equation and simplify.

$$\begin{aligned} y &= 44,505(1.0169)^{10} \\ &\approx 52,625 \end{aligned}$$

About 52,625 visitors are expected 10 years from now. ■

Interpreting Exponential Expressions

You can use the properties of exponents to transform and interpret exponential expressions.

EXAMPLE 3

A deer population with population P is expected to decline according to the model $P(0.344)^{\frac{m}{4}}$, where m is the number of months. What is the monthly rate of decrease?

SOLUTION

Use the properties of exponents to rewrite the expression with only m in the exponent.

$$P(0.344)^{\frac{m}{4}}$$

$$P(0.344)^{\frac{1}{4} \cdot m} \quad \text{Rewrite the exponent.}$$

$$P\left(0.344^{\frac{1}{4}}\right)^m \quad \text{Power of a Power Rule}$$

$$P(0.7658)^m \quad \text{Simplify.}$$

Transform the exponential expression into the form $A(1 + r)^t$ or $A(1 - r)^t$.

$$\begin{aligned} P(0.7658)^m \\ P(1 - 0.2342)^m \end{aligned}$$

The expression can be rewritten in the form $A(1 - r)^t$, where $A = P$ and $r = 0.2342$. Because the rate is subtracted from 1, the expression represents exponential decay.

$$0.2342 = 23.42\% \quad \text{Write the rate as a percent.}$$

The monthly rate of decrease is 23.42%. ■